The sharp constant in Kalenda's Proposition P:schur

arXiv automatic math run `fa_banach_001`

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Source Question

The source paper is Ondrej F. K. Kalenda, *Quantitative Schur property and measures of weak non-compactness*, arXiv:2505.12893 [1]. Its final Question 9.2 asks whether the implication

$$(iii_c) \Longrightarrow (ii_{2c+1})$$

in Proposition P:schur is optimal. The same proposition proves $(i_C) \iff (ii_C)$, so this is equivalently a question about the best constant C in $(iii_c) \Rightarrow (i_C)$.

So, (x_n) is 4-discrete, in particular $\widetilde{\mathrm{ca}}\,(x_n) = 4$. Let us estimate $\delta\,(x_n)$. Let $f \in \mathrm{Lip}_0(M)$ be 1-Lipschitz. Applying the above claim to f and $-f$ we see that for all $n \in \mathbb{N}$ with at most two exceptions we have $|f(n) - f(-n)| \le 1$. In particular, there is $n_0 \in \mathbb{N}$ such that $|f(n) - f(-n)| \le 1$ for $n \ge n_0$. Thus for $m, n \ge n_0$ we have

$$\begin{aligned} |\langle f, x_n - x_m\rangle| &= |f(n) - f(-n) - f(m) + f(-m)| \\ &\le |f(n) - f(-n)| + |f(m) - f(-m)| \le 2. \end{aligned}$$

Thus $\mathrm{ca}\,(\langle f, x_n\rangle) \le 2$. Since f was arbitrary, we deduce $\delta\,(x_n) \le 2$, which completes the proof. □

9. Final remarks and open problems

In this section we collect open problems which arise from the results. Some of them concern optimality of constants and inequalities.

Question 9.1. *Is implication $(iv_c) \to (i_{2c})$ in Proposition 4.1 optimal?*

This question is related to comparison of two natural notion of quantitative weak sequential completeness. We note that we know no example showing that factor 2 is necessary. In fact, we do not know whether some factor greater than 1 is needed. Te next question is similar.

Question 9.2. *Is implication $(iii_c) \to (ii_{2c+1})$ in Proposition 5.1 optimal?*

This question is related to comparison of two natural notions of quantitative Schur property. In this case some increase of constant is necessary, as in Examples 8.3 and 8.4 we provide a Banach space satisfying (iii_1), (ii_2), but not (ii_c) for $c < 2$. But it is not clear whether the optimal constant is $2c + 1$, $2c$ or $c + 1$ (or something else).

A further question is related to the sufficient condition from Section 6.

Question 9.3. *Is the condition from Proposition 6.4 necessary for the 1-Schur property of a real Banach space?*

We note that for complex spaces the condition is not necessary by Remark 6.7(3).

The last question is devoted to Lipschitz-free spaces.

Question 9.4. *Let M be a uniformly separated metric space.*

(1) *Does $\mathcal{F}(M)$ has the quantitative Schur property? Does it have the 3-Schur property?*
(2) *Is $\mathcal{F}(M)$ quantitatively weakly sequentially complete?*
(3) *Are the two measures of non-compactness ($\omega(\cdot)$ an $\mathrm{wk}\,(\cdot)$ equivalent in $\mathcal{F}(M)$?*

As remarked above, due to [1] we know that $\mathcal{F}(M)$ has the Schur property. If M is additionally bounded, we know it has quantitative Schur property by Proposition 8.2. But the key question is whether the constant $\frac{b}{a}$ in the quoted proposition is optimal. Examples 8.3 and 8.5 show that the constant is optimal if $\frac{b}{a}$ equals 2 or 3. But it is not clear how to construct examples with worse constants.

References

[1] Aliaga, R. J., Gartland, C., Petitjean, C., and Procházka, A. Purely 1-unrectifiable metric spaces and locally flat Lipschitz functions. *Trans. Amer. Math. Soc. 375*, 5 (2022), 3529–3567.

Definitions

For a bounded sequence (x_n) in a Banach space X, write

$$\delta(x_n) = \operatorname{diam} \operatorname{clust}_{w^*}(x_n) \subset X^{**}$$

and

$$\widetilde{\operatorname{ca}}(x_n) = \inf\{\operatorname{ca}(x_{n_j}) : n_1 < n_2 < \cdots\}.$$

Condition (iii_c) in Proposition P:schur is

$$\widetilde{\operatorname{ca}}(x_n) \le c\,\delta(x_n) \qquad \text{for every bounded sequence } (x_n) \subset X.$$

Condition (i_C) is

$$\limsup_n \|x_n\| \le C\,\widehat{d}(\operatorname{clust}_{w^*}(x_n), \{0\}) \qquad \text{for every bounded sequence } (x_n) \subset X.$$

Here $\widehat{d}(\operatorname{clust}_{w^*}(x_n), \{0\}) = \sup\{\|x^{**}\| : x^{**} \in \operatorname{clust}_{w^*}(x_n)\}$.

Answer

Theorem 1. *The best constant forced by (iii_c) in Proposition* P:schur *is*

$$\begin{cases} 1, & 0 < c < 1, \\ 2c+1, & c \ge 1. \end{cases}$$

More precisely, if $0 < c < 1$ and X satisfies (iii_c), then X is finite-dimensional and hence satisfies (ii_1). The constant 1 is sharp on nonzero finite-dimensional spaces. If $c \ge 1$, there is a Banach space satisfying (iii_c) and failing (i_C), equivalently (ii_C), for every $C < 2c+1$. Thus Kalenda's implication $(iii_c) \Rightarrow (ii_{2c+1})$ is sharp throughout the nontrivial infinite-dimensional range $c \ge 1$.

The sharp examples for $c \ge 1$

Fix $c \ge 1$, and let $\mathbb{K}$ be either $\mathbb{R}$ or $\mathbb{C}$. On $\mathbb{K} \oplus \ell_1$ define

$$\|(t,x)\|_c = \max\bigl\{\|x\|_1,\ |t| + 2c\|x\|_\infty\bigr\}.$$

This is an equivalent norm on the direct sum; call the resulting Banach space X_c.

Lemma 1 (a gliding-hump estimate)**.** *Let (u_n) be a bounded sequence in ℓ_1 which converges coordinatewise to 0, and suppose that $\|u_n\|_1 \to A$. Then:*

1. *$\delta_{\ell_1}(u_n) \ge 2A$;*
2. *for every $\varepsilon > 0$ there is a subsequence, still denoted (u_n), such that for all $p < q$,*

$$\|u_p - u_q\|_1 \le 2A + \varepsilon, \qquad \|u_p - u_q\|_\infty \le A + \varepsilon.$$

Proof. For the first assertion, fix $\varepsilon > 0$. A standard gliding-hump selection gives a subsequence (v_m) of (u_n) such that

$$\|v_m - v_n\|_1 \geq 2A - \varepsilon \qquad (m \neq n).$$

Indeed, when finitely many earlier vectors have been chosen, choose finite sets carrying almost all of their ℓ_1-mass and then use coordinatewise convergence to make the next vector small on those sets. Hence $\mathrm{ca}(v_m) \geq 2A - \varepsilon$. Since ℓ_1 has the 1-Schur property, $\mathrm{ca}(v_m) \leq \delta_{\ell_1}(v_m)$, and cluster points of a subsequence are cluster points of the original sequence. Therefore $\delta_{\ell_1}(u_n) \geq 2A - \varepsilon$, and $\varepsilon \downarrow 0$ proves the claim.

For the second assertion, discard finitely many terms so that $\|u_n\|_1 \leq A + \varepsilon/4$ for all remaining n. Choose a subsequence inductively as follows. After $u_{n_1}, \ldots, u_{n_m}$ have been chosen, pick a finite set $F_m \subset \mathbb{N}$ with

$$\sum_{j \notin F_m} |u_{n_m}(j)| < \varepsilon/4.$$

Then choose all later indices so large that their coordinates on $\bigcup_{r \leq m} F_r$ have modulus $< \varepsilon/4$. Relabel this subsequence as (u_n). The ℓ_1-estimate is immediate from the triangle inequality. For $p < q$, on F_p the vector u_q is uniformly small, while off F_p the vector u_p is uniformly small. Since $\|u_p\|_\infty, \|u_q\|_\infty \leq A + \varepsilon/4$, this gives $\|u_p - u_q\|_\infty \leq A + \varepsilon$. $\square$

Proposition 1. *For every $c \geq 1$, the space X_c satisfies (iii_c).*

Proof. Let $z_n = (t_n, x_n)$ be a bounded sequence in X_c. To estimate $\widetilde{\mathrm{ca}}(z_n)$, pass to a subsequence such that $t_n \to t$, x_n converges coordinatewise to some $x \in \ell_1$, and

$$\|x_n - x\|_1 \longrightarrow A.$$

The existence of $x \in \ell_1$ follows from Fatou's lemma applied to the coordinatewise limit of the ℓ_1-bounded sequence (x_n).

Set $u_n = x_n - x$. The projection $P : X_c \to \ell_1$, $P(t, x) = x$, has norm at most 1. Therefore

$$\delta_{X_c}(z_n) \geq \delta_{\ell_1}(x_n) = \delta_{\ell_1}(u_n) \geq 2A$$

by Lemma 1: indeed, every weak-star cluster point of (x_n) is the P^{**}-image of a weak-star cluster point of a subnet of (z_n), and P^{**} is a contraction. Now fix $\varepsilon > 0$. Passing to a further subsequence and using Lemma 1, we may suppose that

$$|t_p - t_q| \leq \varepsilon, \qquad \|u_p - u_q\|_1 \leq 2A + \varepsilon, \qquad \|u_p - u_q\|_\infty \leq A + \varepsilon$$

for all $p < q$. Hence

$$\begin{aligned} \|z_p - z_q\|_c &= \max\{\|u_p - u_q\|_1,\ |t_p - t_q| + 2c\|u_p - u_q\|_\infty\} \\ &\leq \max\{2A + \varepsilon,\ \varepsilon + 2c(A + \varepsilon)\}. \end{aligned}$$

Since $c \geq 1$, the right hand side is at most $2cA + (2c+1)\varepsilon$. Thus this further subsequence has Cauchy amplitude at most $2cA + (2c+1)\varepsilon$. Letting $\varepsilon \downarrow 0$ gives

$$\widetilde{\mathrm{ca}}(z_n) \leq 2cA \leq c\,\delta_{X_c}(z_n).$$

Because the original sequence was arbitrary, X_c satisfies (iii_c). $\square$

Proposition 2. *The sequence $y_n = (1, e_n)$ in X_c satisfies*

$$\|y_n\|_c = 2c + 1$$

for every n, while every weak-star cluster point of (y_n) in X_c^{**} has norm exactly 1.

Proof. The norm equality is immediate:

$$\|(1, e_n)\|_c = \max\{1, 1 + 2c\} = 2c + 1.$$

Let $y^{**} \in \mathrm{clust}_{w^*}(y_n)$. The functional $\pi(t, x) = t$ has norm at most 1 on X_c, so $\|y^{**}\| \geq |y^{**}(\pi)| = 1$.

We prove the reverse inequality. Let $f \in X_c^*$ with $\|f\| \leq 1$. Write

$$f(t, x) = s\, t + a(x),$$

where $s \in \mathbb{K}$ and $a \in \ell_\infty = (\ell_1)^*$. Suppose, toward a contradiction, that $|y^{**}(f)| > 1$. Multiplying f by a scalar of modulus one and then passing to infinitely many indices from the cluster subnet, we may find $\eta > 0$ and distinct indices $n_1, n_2, \ldots$ such that

$$\operatorname{Re} f(1, e_{n_j}) > 1 + \eta \qquad (j \in \mathbb{N}).$$

For $m > 2c$, put

$$v_m = \left(1 - \frac{2c}{m},\ \frac{1}{m} \sum_{j=1}^{m} e_{n_j}\right).$$

Then

$$\|v_m\|_c = \max\left\{1,\ 1 - \frac{2c}{m} + \frac{2c}{m}\right\} = 1.$$

Also $|s| \leq 1$, because $\|(1, 0)\|_c = 1$. Therefore

$$\begin{aligned} \operatorname{Re} f(v_m) &= \frac{1}{m} \sum_{j=1}^{m} \operatorname{Re} f(1, e_{n_j}) - \frac{2c}{m} \operatorname{Re} s \\ &> 1 + \eta - \frac{2c}{m}. \end{aligned}$$

For m large this is > 1, contradicting $\|f\| \leq 1$ and $\|v_m\|_c = 1$. Hence $|y^{**}(f)| \leq 1$ for all $\|f\| \leq 1$, so $\|y^{**}\| \leq 1$. $\square$

Proof of sharpness for $c \geq 1$. By Propositions 1 and 2, X_c satisfies (iii_c), while for $y_n = (1, e_n)$

$$\limsup_n \|y_n\|_c = 2c + 1 \quad \text{and} \quad \widehat{d}(\mathrm{clust}_{w^*}(y_n), \{0\}) = 1.$$

Thus (i_C) fails for every $C < 2c+1$. Since Proposition P:schur in the source paper proves $(i_C) \iff (ii_C)$, (ii_C) also fails for every $C < 2c + 1$. The source proposition proves the matching upper implication $(iii_c) \Rightarrow (ii_{2c+1})$, so $2c + 1$ is sharp for $c \geq 1$. $\square$

The small-constant range $0 < c < 1$

Proposition 3. *If $0 < c < 1$ and X satisfies (iii_c), then X is finite-dimensional. Consequently X satisfies (ii_1).*

Proof. The proof of Proposition P:schur in the source paper shows that (iii_c) implies the Schur property. Suppose that X were infinite-dimensional. By Rosenthal's ℓ_1 theorem, together with the Schur property's weak sequential completeness, X contains an isomorphic copy of ℓ_1. James' distortion theorem for ℓ_1 then says that, for every $\varepsilon > 0$, X contains a normalized sequence (x_n) which is $(1-\varepsilon)^{-1}$-equivalent to the unit vector basis of ℓ_1; in particular

$$\left\| \sum_{j=1}^{m} \alpha_j x_j \right\| \geq (1-\varepsilon) \sum_{j=1}^{m} |\alpha_j|$$

for all finitely supported scalar families (α_j). Then every subsequence of (x_n) has Cauchy amplitude at least $2(1-\varepsilon)$, while $\delta(x_n) \leq 2$, since $\|x_n\| = 1$. Applying (iii_c) to (x_n) gives

$$2(1-\varepsilon) \leq \widetilde{\mathrm{ca}}(x_n) \leq c\,\delta(x_n) \leq 2c.$$

Letting $\varepsilon \downarrow 0$ yields $c \geq 1$, a contradiction.

Thus X is finite-dimensional. Every bounded sequence in a finite-dimensional space has a norm-convergent subsequence, so $\widetilde{\mathrm{ca}}(x_n) = 0$ for all bounded sequences. Moreover compactness gives $\mathrm{ca}(x_n) \leq \delta(x_n)$, which is exactly (ii_1). $\square$

Sharpness for $0 < c < 1$. Let X be any nonzero finite-dimensional Banach space. As above, X satisfies (iii_c) for every $c > 0$. Choose $x \in X \setminus \{0\}$ and consider the alternating sequence $x, -x, x, -x, \ldots$. Its Cauchy amplitude is $2\|x\|$, and its weak-star cluster set is $\{x, -x\}$, which has diameter $2\|x\|$. Hence (ii_C) fails for every $C < 1$. Therefore the constant 1 in Proposition 3 is best possible. $\square$

Verification and novelty notes

The arithmetic in the construction is minimal: the test sequence has norm $2c+1$, and the averaging vectors used in Proposition 2 have norm 1. The companion checker script verifies these identities for representative constants.

Local run indexes were searched for arXiv `2505.12893`, `P:schur`, `2c+1`, `iii_c`, and "quantitative Schur". The only matching durable packet was the earlier finite-sum lower-bound partial result from this run. A bounded web search on 2026-06-14 for exact phrases around (iii_c), (ii_{2c+1}), `P:schur`, `2c+1`, and Kalenda found no explicit later solution to Question 9.2.

Human-review recommendation

The main points to check are:

1. the gliding-hump estimate in Lemma 1;
2. the proof that X_c satisfies (iii_c);
3. the averaging argument showing that all weak-star cluster points of $(1, e_n)$ have norm at most 1;
4. the standard Rosenthal–James reduction used only in the range $0 < c < 1$.

References

[1] O. F. K. Kalenda, *Quantitative Schur property and measures of weak non-compactness*, arXiv:2505.12893, 2025.

[2] H. P. Rosenthal, *A characterization of Banach spaces containing ℓ^1*, Proceedings of the National Academy of Sciences of the United States of America 71 (1974), 2411–2413.

[3] R. C. James, *Uniformly non-square Banach spaces*, Annals of Mathematics 80 (1964), 542–550.

Human Review: Sharp Constant for Proposition P:schur

Original automatic proof: `solution_packet.pdf`
Checked date: 15 June 2026
Status: Verified.
Verdict: The proof appears correct. It gives a sharpness example showing that, for $c \geq 1$, the constant $2c+1$ in Kalenda's implication

$$(iii_c) \Longrightarrow (ii_{2c+1})$$

cannot be improved.

Human Comments

The construction is well targeted. For $c \geq 1$, the proof considers

$$X_c = \mathbb{K} \oplus \ell_1$$

with norm

$$\|(t,x)\|_c = \max\{\|x\|_1,\ |t| + 2c\|x\|_\infty\}.$$

The sequence

$$y_n = (1, e_n)$$

then has

$$\|y_n\|_c = 2c+1.$$

On the other hand, the averaging argument shows that every weak-star cluster point of (y_n) in X_c^{**} has norm exactly 1. Hence condition (i_C) fails for every $C < 2c+1$. Since Proposition P:schur in the source paper proves $(i_C) \Longleftrightarrow (ii_C)$, this also gives failure of (ii_C) for every $C < 2c+1$. This proves sharpness in the formulation of Question 9.2.

The proof that X_c satisfies (iii_c) also looks correct. The key point is to take an arbitrary bounded sequence $z_n = (t_n, x_n)$, pass to a subsequence with $t_n \to t$, x_n converging coordinatewise to some $x \in \ell_1$, and $\|x_n - x\|_1 \to A$. The fact that the coordinatewise limit belongs to ℓ_1 follows from Fatou's lemma, after the usual diagonal extraction. The gliding-hump lemma then gives

$$\delta_{\ell_1}(x_n - x) \geq 2A$$

and, after passing to a further subsequence, the estimates

$$\|u_p - u_q\|_1 \leq 2A + \varepsilon, \qquad \|u_p - u_q\|_\infty \leq A + \varepsilon,$$

where $u_n = x_n - x$. These estimates give

$$\widetilde{\mathrm{ca}}(z_n) \leq 2cA \leq c\,\delta_{X_c}(z_n),$$

which is precisely (iii_c).

The only improvement I would suggest is in the proof of Lemma 1. The argument uses the fact that ℓ_1 has the 1-Schur property, namely

$$\mathrm{ca}(w_n) \leq \delta_{\ell_1}(w_n)$$

for bounded sequences in ℓ_1. This is a standard quantitative Schur fact, but it should ideally be cited explicitly. In the source paper, this is recalled in Example 5.9, where ℓ_1 is said to have the 1-Schur property by Kalenda–Spurny, Theorem 1.3. Adding this citation at the use in Lemma 1 would make the proof more self-contained and easier to check.

Aside from this minor citation/reference issue, the main construction, the verification of (iii_c), the cluster-point norm calculation, and the final sharpness argument all look sound.

Review Outcome

Accepted as an essentially correct solution to the optimality question for $c \geq 1$. The small-constant range $0 < c < 1$ is also plausibly handled by the Rosenthal–James argument in the packet, but the main contribution for Question 9.2 is the sharp $2c + 1$ example above.